

Local Computation

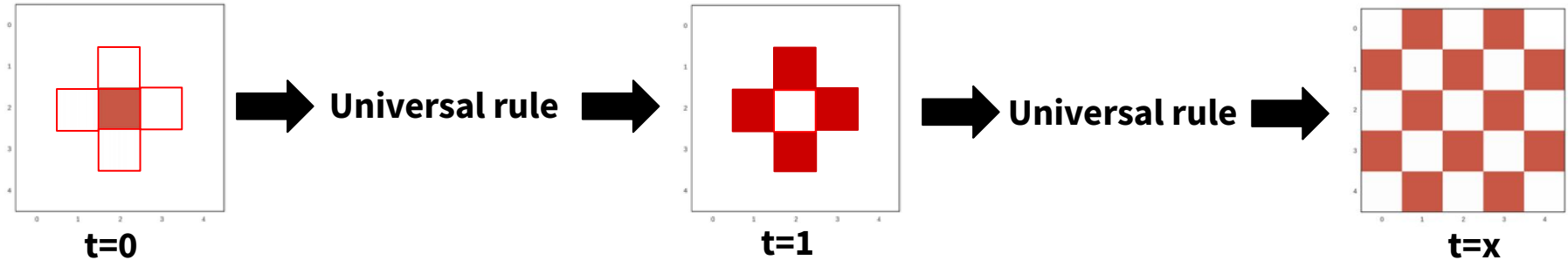
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Motivation



Problem: Manually designing local rules for complex global behaviors is extremely difficult.



However, biology is really good at it.

Zygote



Blastocyst



Embryo



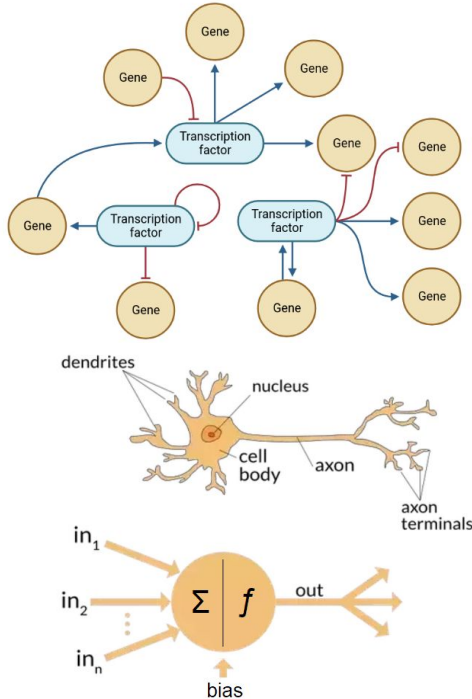
Baby



Biological Inspiration



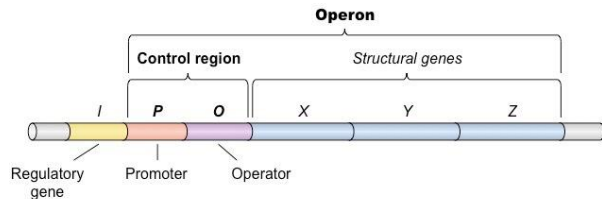
Gene and Neural Networks:



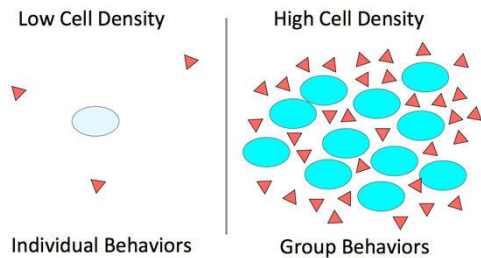
- Networks are akin to weighted network activation or neurons, where one thing activate another
 - Extensively studied in CA
- We use the CA-NEAT framework as our gold standard
 - Demonstrates that NeuroEvolution (NEAT) can discover local rules that result in robust morphogenesis.

Biological Inspiration

Operons, Epigenetics, and Quorum Sensing:



Bacterial Quorum Sensing



- Focus on discrete rules inspired by state based biological mechanisms
 - Akin to evolving Push programs
- Experimentation with evolving Push based cellular automata is underrepresented

Objective



Evaluate whether stack-based genetic programming (Push) offers advantages over neural cellular automata for evolving CA update rules, using self-repair as a case study of emergence.



Push Cellular Automata (PCA)

- A flat, **linear sequence** of terms, all drawn from a predefined set
 - **Logic gates:** *and*, *or*, *xor*, etc
 - **Neighborhood Values:** $N0$ through Nn
 - **Constants:** *true* and *false*
- At the end of execution the top of the *:bool* stack is the cell's next state
- UMAD is typically applied as mutation



Neural Cellular Automata (NCA)

- **Compositional Pattern Producing Networks (CPPN)** are directed acyclic graphs represented as a Clojure map
- Every generation, mutation is applied to the weights, and architecture
 - **Weight Mutation:** 90% are perturbed by noise, 10% fully replaced by random value in range $[-2, 2]$
 - **Add Connection:** Two unconnected nodes connected with random weight
 - **Add Node:** Picks a random connection, splits it, disables the original path and adds a random activation function



Neural Cellular Automata (NCA)

```
{:nodes      {1 {:type :input,  :activation :sigmoid}
               2 {:type :hidden, :activation :tanh}
               3 {:type :output, :activation :sigmoid}}

:connections [{:in 1 :out 2 :weight 0.42 :enabled true}
               {:in 2 :out 3 :weight -1.10 :enabled true}]

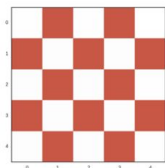
:eval-order  [1 2 3]}
```



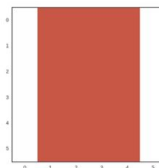
CA-NEAT Benchmark Comparison

Implementation	CA-Neat	Push
Neighborhood	Von Neumann/Cardinal	Von Neumann/Cardinal
Time Steps	30	30
Error	Min. over all time steps	Min. over all time steps
Mutation	Point mutation, Crossover	UMAD
Selection	Species based	Elitism, Tournament (s=5)
Generations	Until Completion	25 - 100

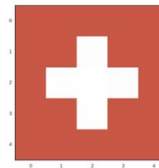
Target Patterns:



(a)



(b)

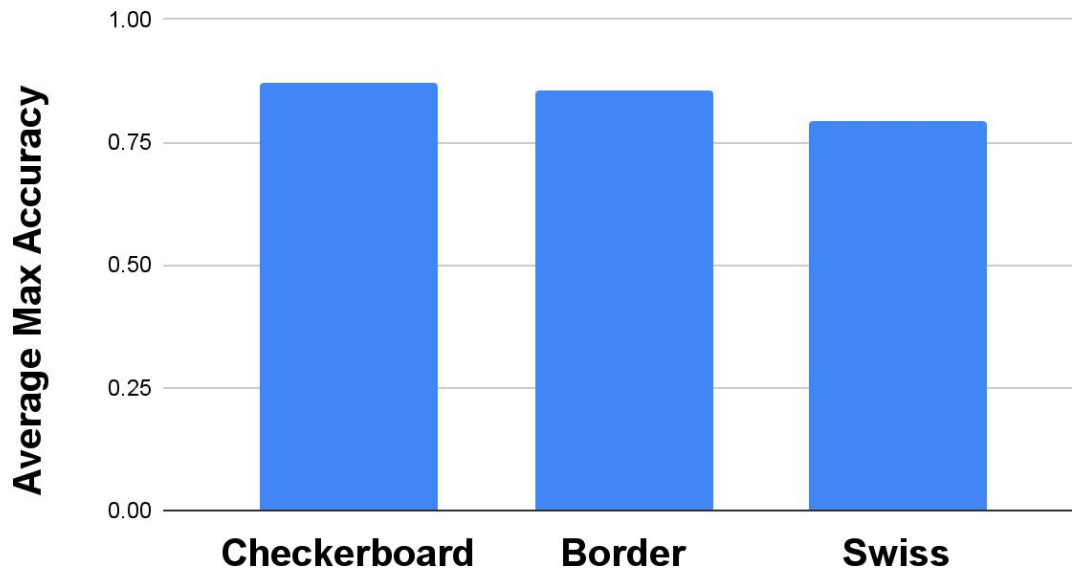


(d)



CA-NEAT Benchmark Comparison

Push Program Performance on CA-Neat Benchmarks

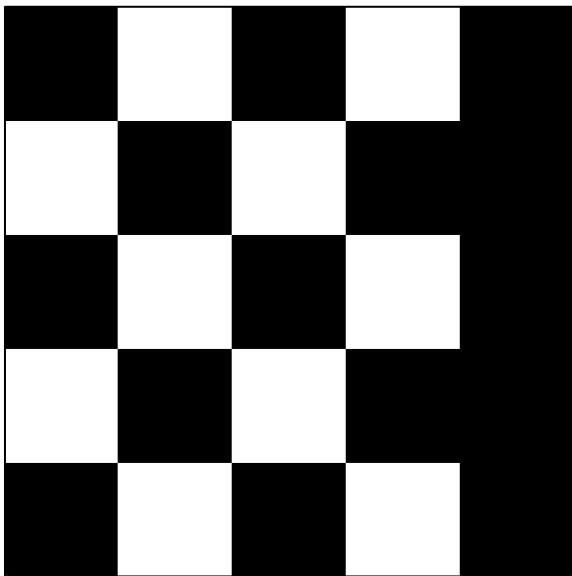


CA-Neat Results

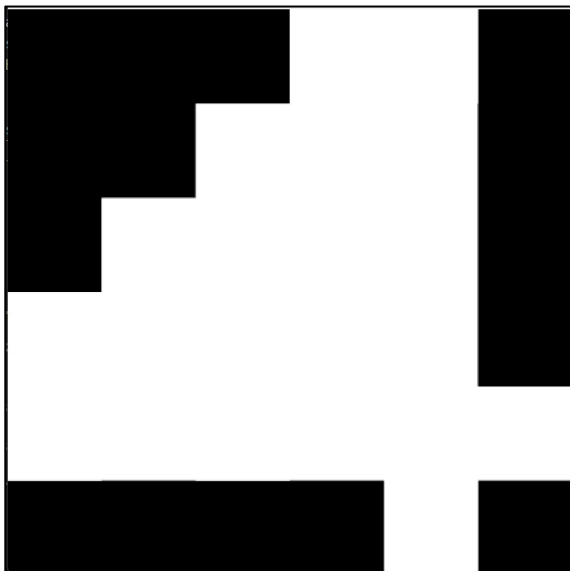
Problem	Success rate %
Mosaic morphogenesis	100
Border morphogenesis	1
Tricolor morphogenesis	100
Swiss morphogenesis	76

CA-NEAT Benchmark Comparison

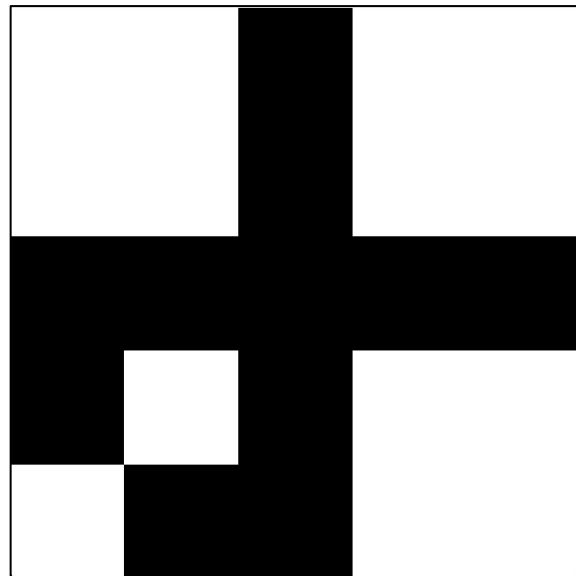
Checkerboard



Border

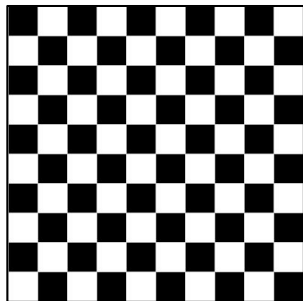


Swiss

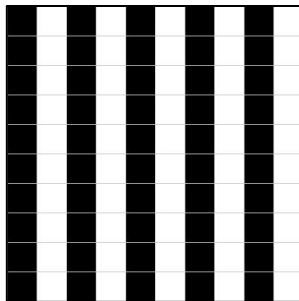


Baseline PCA vs NCA (Targets)

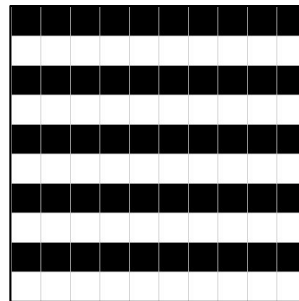
Checkerboard



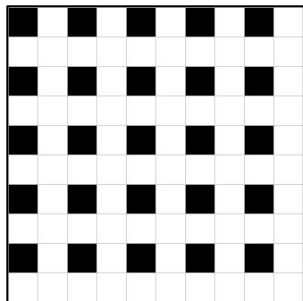
Vertical Stripes



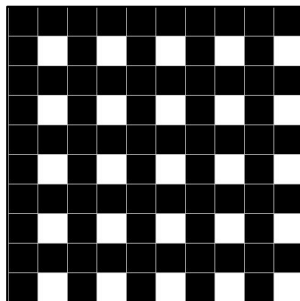
Horizontal Stripes



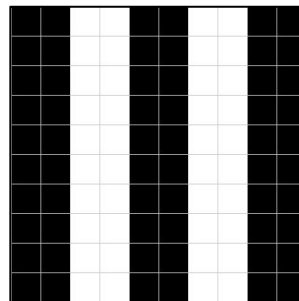
Sparse Dots



Dense Dots



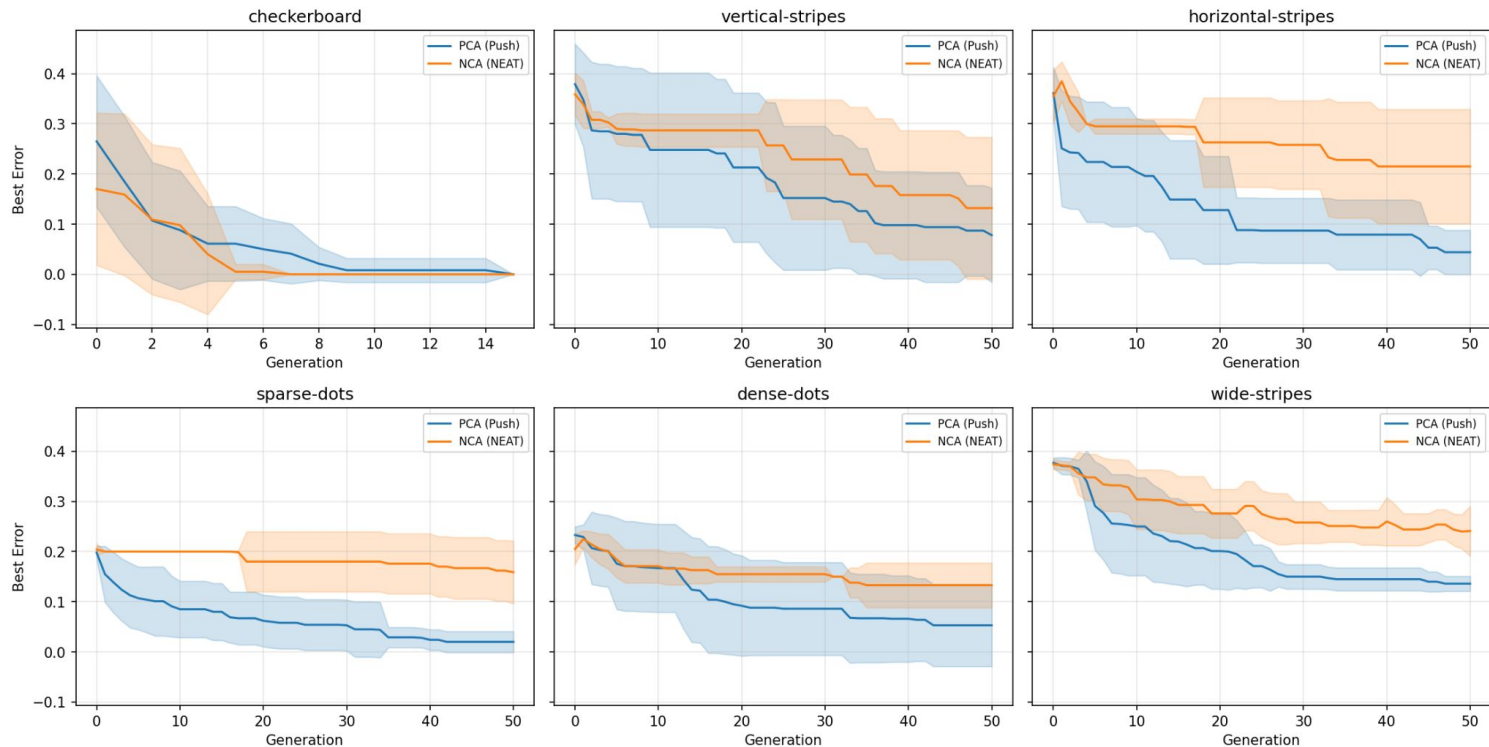
Wide Stripes





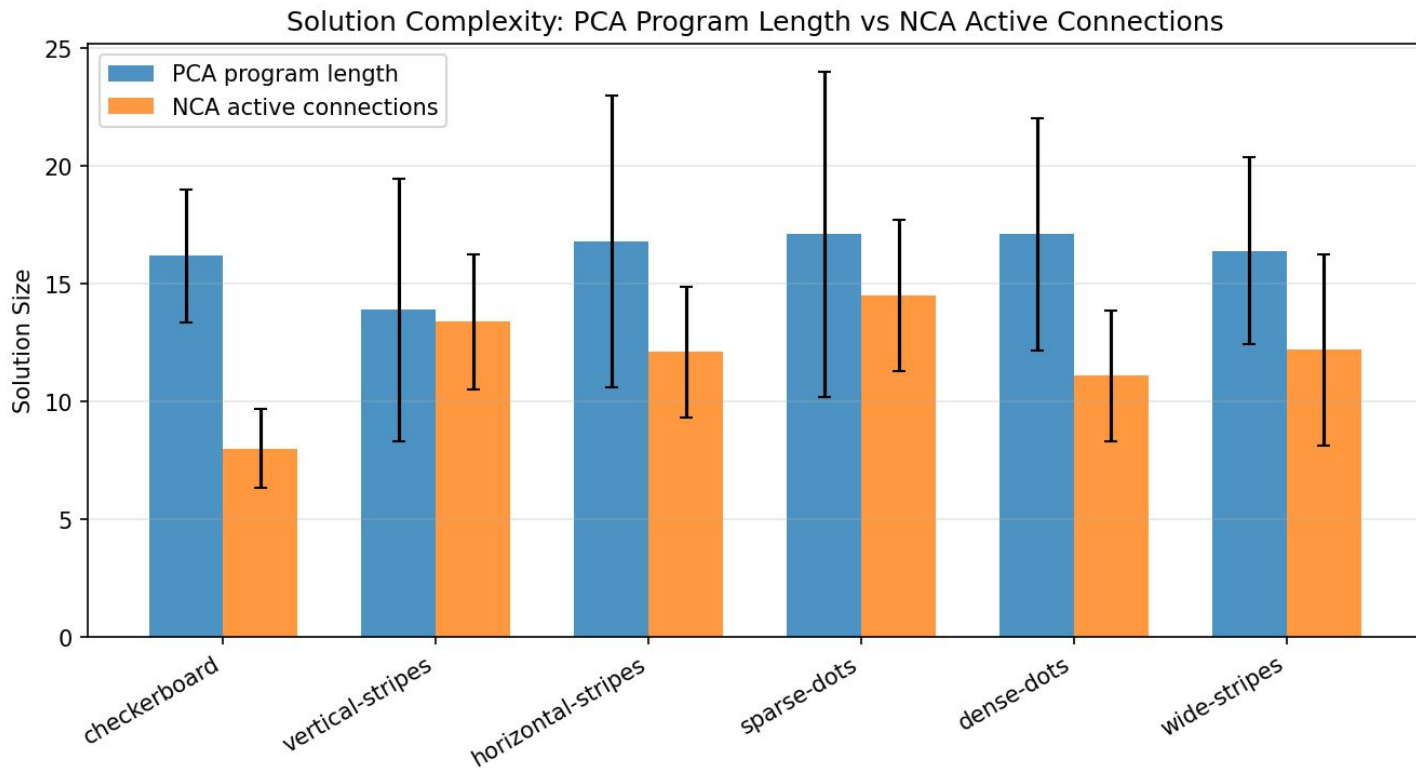
Baseline PCA vs NCA (Convergence)

Error Convergence: Locally-Solvable Patterns



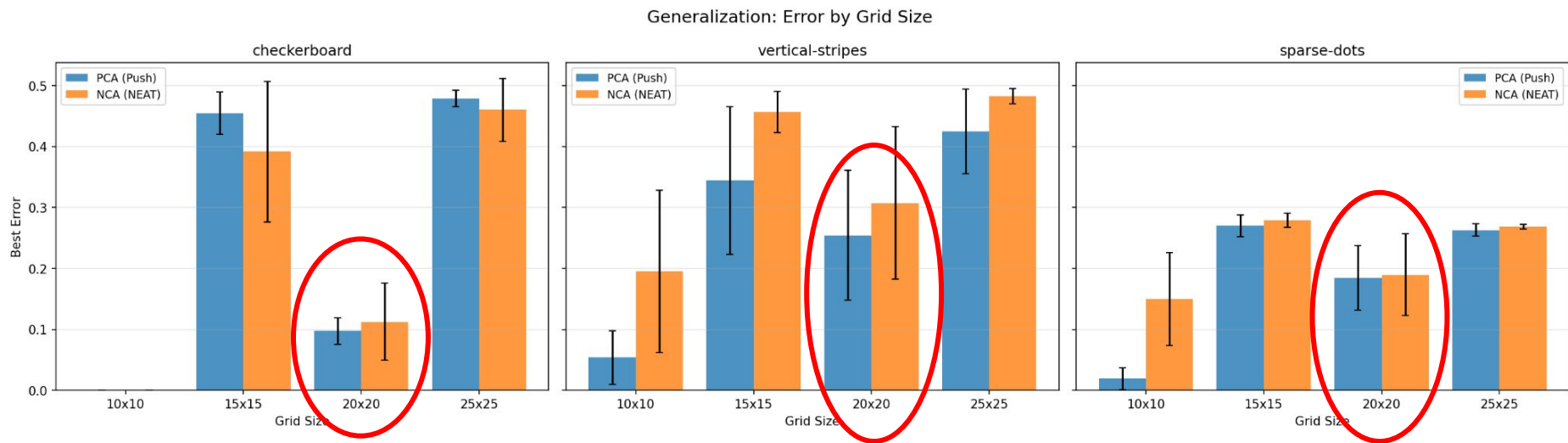


Baseline PCA vs NCA (Complexity)





Rule Generalizability





Emergence of Self-Repair

- Work-in-progress!
 - Current prototype evolves Push local rules under random deletions
 - We are able to successfully evolve rules in this system
- Self-repair of uniform patterns is uninteresting...



Implications & Next Steps

- **Implications:** In its current state PCA is competitive when compared against the baseline NCA
 - However, we still fail to match the effectiveness of CA-NEAT
- **Next Steps:** Biological cells often use a combination of both state-based and continuous network mechanisms to define their local rules
 - Ex: an external chemical signal activates a gene network
- Can we reproduce this framework by using a neural network to selectively activate different Push programs?

Thank you!